Daksh Singh

Bengaluru, India · mailtodakshsingh@gmail.com
Portfolio: portfolio-website-gamma-ivory-77.vercel.app · Team: linkedin.com/company/the-orcas

EDUCATION

Inventure Academy, Bengaluru, Cambridge curriculum — Class of 2027

- **Current:** A Level Mathematics, Physics, Chemistry, Computer Science; AS English Language.
- **AS Level (2026):** Physics A, Chemistry A, Computer Science A, Mathematics B (retake Oct/Nov 2026)
- **IGCSE:** A* in Mathematics (Extended), Physics, Chemistry, Computer Science; 3 A, 1 B; ICE Distinction
- **SAT:** 1500 (superscore; Math 800, Reading & Writing 700) · **AP Computer Science A:** 5

ENGINEERING & RESEARCH

MATE ROV, Bangalore Bullsharks (2025), The Orcas (2026) / *CTO (2026)* — 2025–26

- First Indian team to compete in the Ranger class at the MATE ROV World Championship (2025). Built The Orca (2026) in about two months for ~$600, against rival budgets of $15,000–20,000.
- Led arm design: designed and built The Orca's single-motor gripper; Software Lead on the 2025 ROVs.

BLADE: Botanical Leaf Analysis & Disease Evaluator / *Lead Engineer & Co-researcher* — 2026

- Built a handheld, fully offline crop-disease scanner (Raspberry Pi 4B, 5 MP camera, touchscreen, 3D-printed enclosure); a pretrained MobileNetV2 classifies leaf photos into 38 classes across 14 crops.
- Designed a controlled experiment that photographs the same leaves while changing one condition at a time, to test why lab-trained models fail in real fields. Paper under review.

"Which House Plants Actually Improve Indoor Air Quality?" / *First author, Zenodo preprint* — 2024–25

- Measured CO2, formaldehyde and VOC in five rooms after adding plants: only small reductions.

WRO Future Engineers (self-driving car) / *Team Lead (2025); Team Member (2024)* — 2024–25

- Placed 19th of 40+ at the 2025 national championship. Built self-driving cars with Raspberry Pi, OpenCV and Ackermann steering, including a custom differential from a disassembled RC truck.

National Robotics League (NRL) / *Team Captain (2025, 2026)* — 2025–26

- Placed 20th of 50 in division (2025). Designed the autonomous mode: ESP32, infrared sensors, PID control.

Roboprenr Lab / *Lab Student* — 2024–26

- 5 hands-on lab courses completed, A+ in all, from electronics to AI robotics; ROS 2 in progress.

LEADERSHIP & SERVICE

Tech for Change Robotics Club, Inventure Academy / *Founder & Instructor* — 2025–present

- Founded the school's first senior robotics club, one of two of its 20+ clubs run by a student; grew from 5 to 10 students (signup cap). Built a voting machine that will run the 2027 student council elections.

Student Council / *House Sports Captain & Beyond Academics Captain, Adventurers House* — 2026–27

- Co-ran swim meet (200 athletes; house won 1st), parent-teacher day (350), Tournament of Minds (500).

Project Vita / *Co-founder; On-Site Server & Social Media Support* — 2023–present

- Ran 9+ food stalls raising ~$10,000 for 5 causes, including CareForAll medical camps for ~750 children.

HONORS & AWARDS

Sharkpedo Legacy Award, MATE ROV World Championship (sole recipient, ~120 teams) — 2026
US Navy Challenge Coin, MATE ROV World Championship (~4 of ~120 teams) — 2025
1st place, MATE ROV India Nationals (national record 280/375 in 2026) — 2025, 2026
Harvard Crimson Global Quiz Bowl, Computer Science (High School): Global Third Place, Team — 2025
iCode Global Hackathon (Python): national finalist — 2025

Technical Skills: Python, OpenCV, Raspberry Pi, ESP32, Pixhawk/ArduSub, Onshape, PCB design, 3D printing